

Undecidable Problems for Context-free Languages

The following problems relating to context-free languages are undecidable:

1. How can we tell whether or not two different CFG's define the same language?
2. Given a particular CFG, how can we tell whether or not it is ambiguous?
3. Given a CFG, how can we tell whether or not it has an equivalent PDA that is deterministic?
4. Given a CFG that is ambiguous, how can we tell whether or not there is a different CFG that generates the same language but is not ambiguous?
5. How can we tell whether or not the complement of a CFL is also context-free?
6. How can we tell whether or not the intersection of 2 CFL's is also context-free?
7. Given 2 CFG's, how can we tell whether or not they have a string in common?
8. Given a CFG, how can we tell whether or not there are any strings that it *does not* generate? (Is its language all of $(a + b)^*$ or not?)